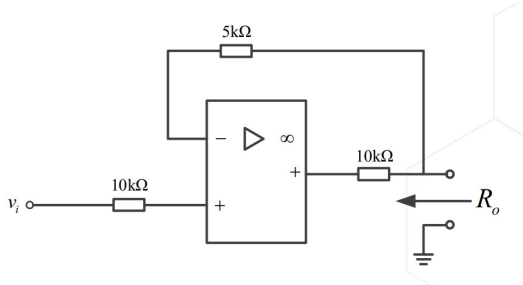
第二次作业

1、考虑改进的运放模型 G=10^5，R_i=1MΩ，R_{cm}=200mΩ，解出图 P1 中电阻 R_o。



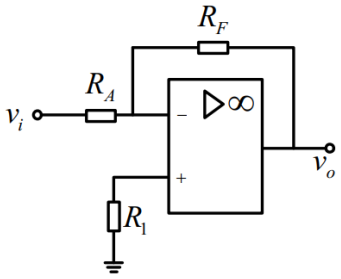
该电路为同相放大电路，由图中所示，套用公式计算如下：

$$R_A=\infty \quad R_1=10k\Omega \quad R_F=5k\Omega \quad R_o^{'}=10k\Omega$$

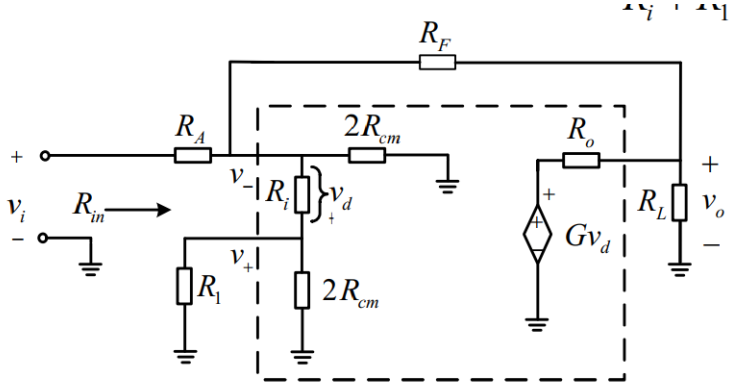
$$R_{out}=\frac{R_o^{'}}{G_0\beta}=\frac{R_o^{'}}{G_0}(1+\frac{R_F}{R_A})=\frac{10k\Omega}{10^5}(1+\frac{5k\Omega}{\infty})=0.1\Omega$$

具体推导过程参阅 PPT 中同相放大电路输出电阻推导过程，不同之处是该改进的电路在原图中 R_o 位置处串接了 10kΩ 的电阻，10kΩ 远远大于原 R_o，等效为 10kΩ 即可，其余推导过程均一致。

2、基于等效电路模型，推导反相放大电路的增益，输入电阻，输出电阻和带宽。



（1）推导如下，反相放大电路等效电路模型为：



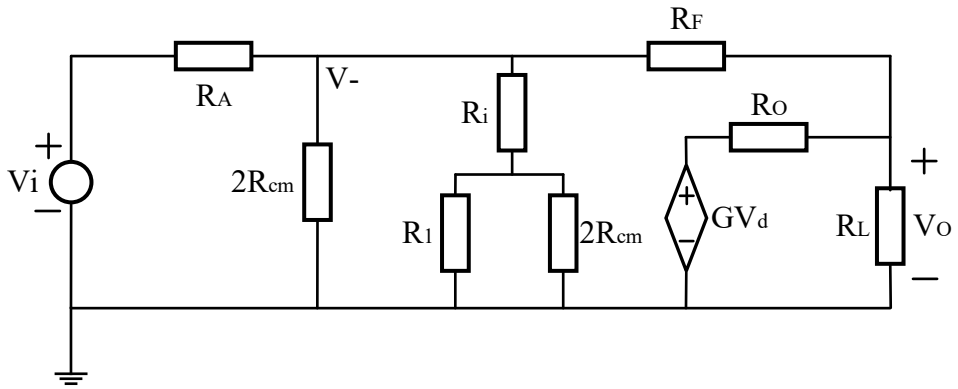
$$R_1^{'}=R_1 \,||\, 2R_{cm}$$

$$R_F >> R_O$$

$$R_L >> R_O$$

$$v_d = -\frac{R_i v_-}{R_i + R_1'}$$

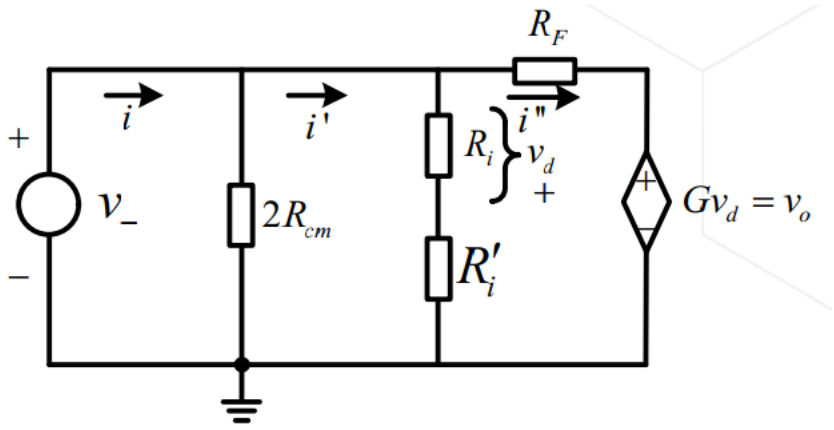
进一步等效：



由上图可得：

$$v_O = \frac{R_L}{R_L + R_O} G V_d \approx G V_d$$

进一步等效：



由该图得出：

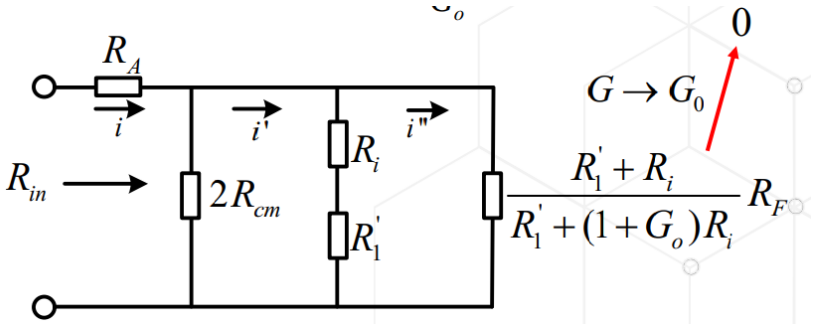
$$R_F i'' = v_- - G V_d = v_- + \frac{G v_-}{R_i + R_1'} = \frac{R_i(1+G) + R_1'}{R_i + R_1'} v_-$$

则等效电阻为：

$$\frac{v_-}{i''} = \frac{(R_i + R_1')R_F}{R_i(1+G) + R_1'}$$

G0 很大，该值趋于 0

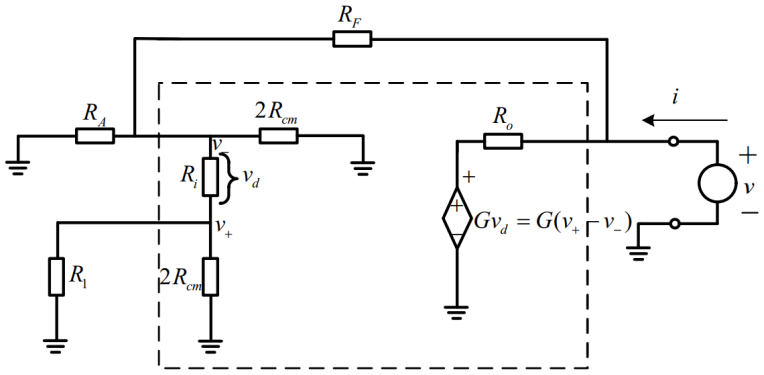
再次等效：



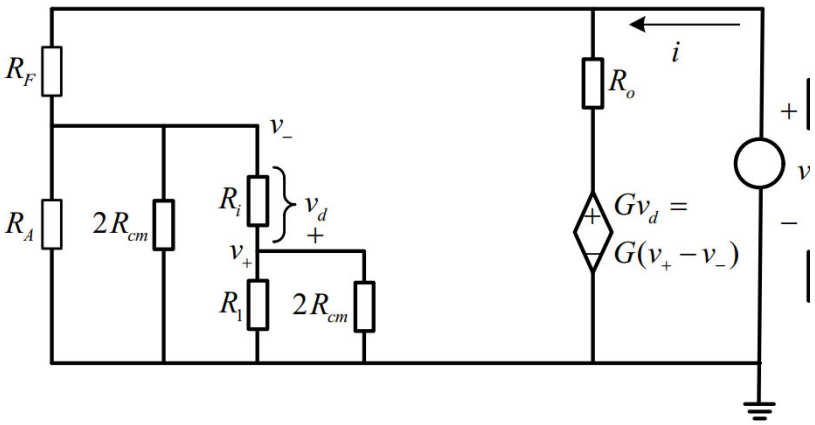
因此输入电阻为：

$$R_{in} = R_A + 2R_{cm} \,||\, (R_i + R_1') \approx R_A$$

（2）推导输出电阻，由于都是将输入端置 0，与同相放大电路的输出电阻推导完全一致：

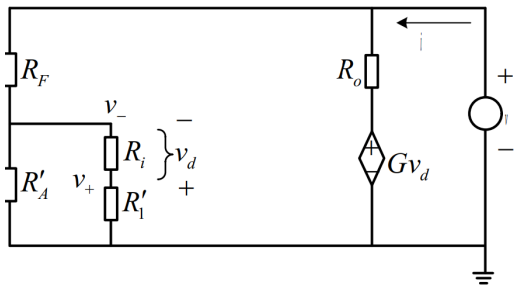


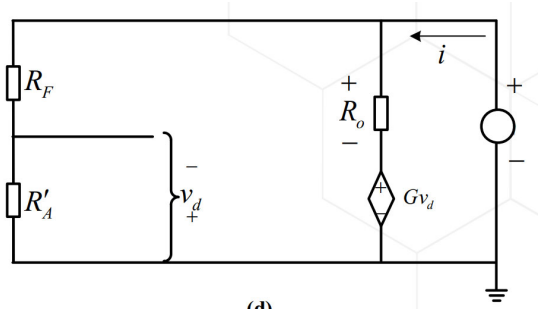
将电路做第一次等效：



$$R_1' = R_1 \,||\, 2R_{cm} \approx R_1$$

$$R_A' = R_A \,||\, 2R_{cm} \approx R_A$$





(d)

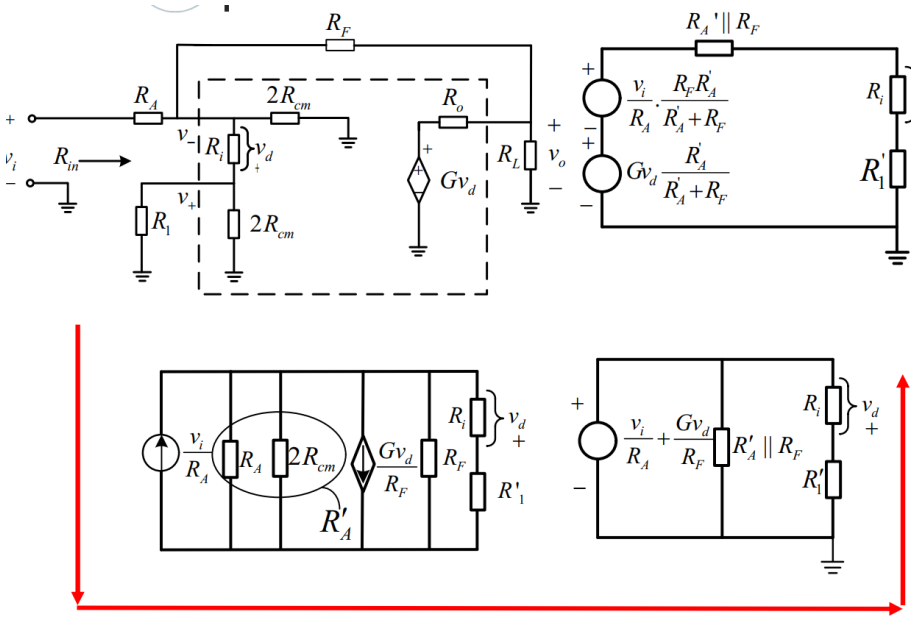
$$v_d = -\frac{R_A^{'} v}{R_A^{'} + R_F}$$

$$iR_O = v - GV_d = v(1 + \frac{GR_A^{'}}{R_A^{'} + R_F})$$

$$R_{out} = \frac{v}{i} = \frac{R_O}{1 + GR_A^{'} / (R_A^{'} + R_F)}$$

$$R_{out} = \frac{R_O}{1 + G_0\beta} \approx \frac{R_O}{G_0\beta}$$

（3）推导电压增益



推导电压增益的过程中使用了戴维南定理以及诺顿定理进行电路等效，将电路简化：

$$v_d = -\frac{R_i}{R_i + R_1^{'} + R_A^{'} \| R_F}[(R_A^{'} \| R_F)\frac{v_i}{R_A} + \frac{R_A^{'}G}{R_A^{'} + R_F} v_d]$$

因 $v_O = Gv_d$，电压增益为：

$$A_{-} = \frac{v_O}{v_i} = -\frac{GR_A^{'}R_i R_F / R_A}{(R_i + R_1^{'})(R_A^{'} + R_F) + R_A^{'}R_F + R_i R_A^{'}G}$$

$$R_A << 2R_{cm}，\ G>>1$$

$$R_A^{'} = R_A \| 2R_{cm} = R_A$$

$$A_{-} = -\frac{R_F}{R_A}\frac{1}{1 + 1/G(R_A/(R_A + R_F))} = -\frac{R_F}{R_A}\frac{1}{1 + 1/G\beta}$$

推导出：

$$A_{-} = -\frac{R_F}{R_A}$$

（4）带宽的推导：

$$R_A^{'} = R_A，\ R_i >> R_1^{'}，\ R_i >> R_F$$

$$G = \frac{G_0}{1 + s/\omega_0}$$

$$A_{-} = -\frac{R_F}{R_A}\frac{G_0/(1 + s/\omega_0)}{(1 + R_F/R_A) + G_0/(1 + s/\omega_0)} = -\frac{R_F}{R_A}\frac{G_0}{(1 + R_F/R_A)(1 + s/\omega_0) + G_0}$$

由于 $G_0 >> (1 + R_F/R_A)$

$$A_{-} \approx -\frac{R_F}{R_A}\frac{1}{(1 + R_F/R_A)s/G_0\omega_0 + 1}$$

$$\omega_0 = BW = \frac{G_0\omega_0}{1 + R_F/R_A} = \frac{GBP}{1 + R_F/R_A} = \beta \times GBP$$